GEOMETRIC NORMALITY SATISFIES THE FIVE PROPERTIES REQUIRED OF A FORMAL FIBRE CONDITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. The author's proof is a checklist: one line per property, each naming the exact result it follows from (definition, local characterisation, ascent, descent, separable algebraic base change). Terse but complete, not a hint-only proof.

Allowed standard prerequisites (editorial).

- Definition of geometric normality and ascent of normality. Algebra, Definition \ref{algebra-definition-geometrically-normal} and Lemma \ref{lemma-normal-goes-up}, cited at proof lines 13221 and 13225. Give properties (A) and (C).
- Normality descends along ring maps, and geometric normality is preserved by separable algebraic base change. Algebra, Lemma \ref{algebra-lemma-descent-normal} and Algebra, Lemma \ref{algebra-lemma-geometrically-normal-over-separable-algebraic}, cited at proof lines 13227-13229. Give properties (D) and (E).

Source reference. Original lemma 0BIX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14505–14505. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0380: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4219–4219. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0BFK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14388–14388. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 033G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4055–4055. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0C31: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15176–15176. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Properties (A), (B), (C), (D), and (E) hold for $P(k \to R) =$"R is geometrically normal over k".*

Proof. Part (A) follows from the definition of geometrically normal algebras (Algebra, Definition 0380). Part (B) follows too: a ring is normal if and only if all of its local rings are normal. Part (C). This follows from Lemma 0BFK. Part (D). This follows from Algebra, Lemma 033G. Part (E). This follows from Algebra, Lemma 0C31. □